

Numerical Studies of Cooling in the Spin Model

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I. EXACT DIAGONALIZATION STUDY OF COOLING IN THE SPIN MODEL

We analyze cooling protocols for fermionic systems and connect them to spin systems via the Jordan-Wigner (JW) transformation. This enables a similar analysis for spin systems, which are more directly implementable in many experimental platforms.

Mapping both the system and bath together presents a challenge: the JW transformation introduces non-local string operators, rendering the combined system non-integrable. Therefore, we apply the transformation only to the system Hamiltonian, which allows us to compute the mode energies ϵ_k and track which modes are resonating and being cooled during the protocol.

A. The Transverse Field Ising Model

We study a related model of spins

$$H_{\text{spin}} = \frac{\cos \theta}{2} \left(\sum_{n=1}^{N-1} \sigma_{n,x} \sigma_{n+1,x} + g_I \sigma_{N,x} \sigma_{1,x} \right) + \frac{\sin \theta}{2} \sum_{n=1}^N \sigma_{n,z}, \quad (1)$$

where $\sigma_{n,\alpha}$ ($\alpha = x, y, z$) are the Pauli operators acting on site n , and N is the number of spins (assumed to be even). The boundary condition is determined by the parameter g_I : $g_I = +1$ for periodic boundary conditions (PBC), $g_I = -1$ for anti-periodic boundary conditions (APBC), and $g_I = 0$ for open boundary conditions (OBC).

B. Jordan-Wigner Transformation

To map the spin model to fermions, we employ the JW transformation using the string operator $S_n = \prod_{j=1}^{n-1} \sigma_{j,z}$. The fermionic creation and annihilation operators are defined as:

$$a_n = -S_n \cdot \frac{\sigma_{n,x} - i\sigma_{n,y}}{2}, \quad (2)$$

$$a_n^\dagger = -S_n \cdot \frac{\sigma_{n,x} + i\sigma_{n,y}}{2}. \quad (3)$$

These operators satisfy the standard fermionic anti-commutation relations $\{a_m, a_n\} = \{a_m^\dagger, a_n^\dagger\} = 0$ and $\{a_m, a_n^\dagger\} = \delta_{mn}$.

Inverting the transformation, we can express the Pauli operators in terms of fermions:

$$\sigma_{n,x} = -S_n(a_n + a_n^\dagger), \quad (4)$$

$$\sigma_{n,y} = -iS_n(a_n - a_n^\dagger), \quad (5)$$

$$\sigma_{n,z} = 2a_n^\dagger a_n - 1 = -(a_n a_n^\dagger - a_n^\dagger a_n). \quad (6)$$

C. Transforming the Hamiltonian

We now apply the JW transformation to each term in the Hamiltonian.

For the transverse field term, substituting $\sigma_{n,z} = 2a_n^\dagger a_n - 1$, we obtain:

$$\begin{aligned} \frac{\sin \theta}{2} \sum_{n=1}^N \sigma_{n,z} &= \frac{\sin \theta}{2} \sum_{n=1}^N (2a_n^\dagger a_n - 1) \\ &= -\frac{\sin \theta}{2} \sum_{n=1}^N (a_n a_n^\dagger - a_n^\dagger a_n) \end{aligned} \quad (7)$$

For the nearest-neighbor interaction terms in the bulk, $\sigma_{n,x} \sigma_{n+1,x}$, we start from the fermionic representation of $\sigma_{n,x}$:

$$\sigma_{n,x} \sigma_{n+1,x} = [-S_n(a_n + a_n^\dagger)] [-S_{n+1}(a_{n+1} + a_{n+1}^\dagger)] \quad (8)$$

$$= S_n(a_n + a_n^\dagger) S_{n+1}(a_{n+1} + a_{n+1}^\dagger). \quad (9)$$

Using the identity $S_{n+1} = S_n \sigma_{n,z}$ and $S_n^2 = \mathbb{1}$, we have

$$\begin{aligned} \sigma_{n,x} \sigma_{n+1,x} &= S_n(a_n + a_n^\dagger) S_n \sigma_{n,z} (a_{n+1} + a_{n+1}^\dagger) \\ &= (a_n + a_n^\dagger) \sigma_{n,z} (a_{n+1} + a_{n+1}^\dagger), \end{aligned} \quad (10)$$

where we used $[S_n, a_n] = 0$ and $S_n^2 = \mathbb{1}$. Using $\sigma_{n,z} = 2a_n^\dagger a_n - 1$, one verifies the local identity

$$(a_n + a_n^\dagger) \sigma_{n,z} = a_n - a_n^\dagger. \quad (11)$$

Therefore,

$$\begin{aligned} \sigma_{n,x} \sigma_{n+1,x} &= (a_n - a_n^\dagger)(a_{n+1} + a_{n+1}^\dagger) \\ &= a_n a_{n+1} + a_n a_{n+1}^\dagger - a_n^\dagger a_{n+1} - a_n^\dagger a_{n+1}^\dagger. \end{aligned} \quad (12)$$

The boundary term requires special attention due to the non-local nature of the JW strings. For a closed chain ($g_I = \pm 1$), the fermionic Hilbert space splits into two sectors based on fermion parity. The parity operator is defined as

$$P = \prod_{j=1}^N \sigma_{j,z} = \prod_{j=1}^N (2a_j^\dagger a_j - 1) = (-1)^{\sum_{j=1}^N n_j} = (-1)^{N_f} \quad (13)$$

where $N_f = \sum_{j=1}^N n_j$ is the total fermion number. For the boundary term, we can show that $S_N S_1 = P \cdot \sigma_{N,z}$, where P is the parity operator (since $S_N = \prod_{j=1}^{N-1} \sigma_{j,z} = \prod_{j=1}^N \sigma_{j,z} \cdot \sigma_{N,z}$ and $S_1 = \mathbb{1}$). Therefore:

$$\sigma_{N,x} \sigma_{1,x} = -P \cdot (a_N - a_N^\dagger)(a_1 + a_1^\dagger) \quad (14)$$

$$= -P \cdot (a_N a_1 + a_N a_1^\dagger - a_N^\dagger a_1 - a_N^\dagger a_1^\dagger). \quad (15)$$

In the even parity sector ($P = 1$):

$$\sigma_{N,x}\sigma_{1,x} = -(a_N a_1 + a_N a_1^\dagger - a_N^\dagger a_1 - a_N^\dagger a_1^\dagger). \quad (16)$$

In the odd parity sector ($P = -1$):

$$\sigma_{N,x}\sigma_{1,x} = a_N a_1 + a_N a_1^\dagger - a_N^\dagger a_1 - a_N^\dagger a_1^\dagger. \quad (17)$$

Combining all terms, the transformed Hamiltonian in the even/odd parity sector is:

$$H_{\text{spin}}^{e/o} = \frac{\cos \theta}{2} \left[\sum_{n=1}^{N-1} (a_n - a_n^\dagger)(a_{n+1} + a_{n+1}^\dagger) - g_I P (a_N - a_N^\dagger)(a_1 + a_1^\dagger) \right] - \frac{\sin \theta}{2} \sum_{n=1}^N (a_n a_n^\dagger - a_n^\dagger a_n) \quad (18)$$

$$= \frac{\cos \theta}{2} \left[\sum_{n=1}^{N-1} (a_n a_{n+1} + a_n a_{n+1}^\dagger - a_n^\dagger a_{n+1} - a_n^\dagger a_{n+1}^\dagger) - g_I P (a_N a_1 + a_N a_1^\dagger - a_N^\dagger a_1 - a_N^\dagger a_1^\dagger) \right] - \frac{\sin \theta}{2} \sum_{n=1}^N (a_n a_n^\dagger - a_n^\dagger a_n) \quad (19)$$

where P is the fermion parity, which is $+1$ in the even sector and -1 in the odd sector.

To emphasize the role of the boundary condition in the fermionic problem, we introduce $g_F = -g_I \cdot P$, which gives:

$$H_{\text{spin}}^{e/o} = \frac{\cos \theta}{2} \left[\sum_{n=1}^{N-1} (a_n a_{n+1} + a_n a_{n+1}^\dagger - a_n^\dagger a_{n+1} - a_n^\dagger a_{n+1}^\dagger) + g_F (a_N a_1 + a_N a_1^\dagger - a_N^\dagger a_1 - a_N^\dagger a_1^\dagger) \right] - \frac{\sin \theta}{2} \sum_{n=1}^N (a_n a_n^\dagger - a_n^\dagger a_n) \quad (20)$$

Equivalently, in a fixed parity sector (so g_F is a fixed number), we can write the interaction term in a translation-invariant form by defining the fermionic boundary condition

$$a_{N+1} = g_F a_1, \quad a_{N+1}^\dagger = g_F a_1^\dagger. \quad (21)$$

Then the Hamiltonian can be expressed compactly as

$$H_{\text{spin}}^{g_F} = \frac{\cos \theta}{2} \sum_{n=1}^N (a_n - a_n^\dagger)(a_{n+1} + a_{n+1}^\dagger) - \frac{\sin \theta}{2} \sum_{n=1}^N (a_n a_n^\dagger - a_n^\dagger a_n) \quad (22)$$

where a_{N+1} is understood via the boundary condition above. This is the starting point for the Fourier/Bogoliubov diagonalization below.

For a spin system with periodic boundary conditions ($g_I = 1$), the fermionic representation has $g_F = -1$ in the even parity sector and $g_F = +1$ in the odd parity sector, corresponding to antiperiodic and periodic boundary conditions for the fermions, respectively. For a spin system with anti-periodic boundary conditions ($g_I = -1$), this is swapped: $g_F = +1$ in the even parity sector and $g_F = -1$ in the odd parity sector.

D. Fourier Transformation

To diagonalize the quadratic fermionic Hamiltonian, we define $\phi_k = \frac{2\pi k}{N}$ and transform to momentum space using the Fourier transform:

$$\tilde{a}_k = \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{-in\phi_k} a_n, \quad (23)$$

$$\tilde{a}_k^\dagger = \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{+in\phi_k} a_n^\dagger. \quad (24)$$

The inverse transformations are:

$$a_n = \frac{1}{\sqrt{N}} \sum_k e^{+in\phi_k} \tilde{a}_k, \quad (25)$$

$$a_n^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-in\phi_k} \tilde{a}_k^\dagger. \quad (26)$$

This is the standard “origin at $n = 0$ ” Fourier convention written on the site labels $n = 1, \dots, N$. Using $e^{\pm i(n-1)\phi_k}$ instead (common when emphasizing the site index $n = 1$ as the origin) would simply redefine the momentum operators by a k -dependent phase ($\tilde{a}_k^{(n-1)} = e^{+i\phi_k} \tilde{a}_k^{(n)}$), and does not affect any physical results as long as the convention is used consistently. Here, the sum over k is determined by the fermionic boundary conditions, which depend on the spin parity sector. For periodic boundary conditions ($g_F = 1$), k takes integer values $k \in \{-\frac{N}{2} + 1, \dots, \frac{N}{2}\}$. For antiperiodic boundary conditions ($g_F = -1$), k takes half-integer values $k \in \{-\frac{N-1}{2}, \dots, \frac{N-1}{2}\}$.

E. Fermionic PBC $g_F = 1$ Case

For $g_F = +1$ (fermionic PBC) the allowed momenta are integers $k \in \{-\frac{N}{2} + 1, \dots, \frac{N}{2}\}$ (so $\phi_k = \frac{2\pi k}{N}$ includes the special values $\phi = 0$ and $\phi = \pi$). Using the Fourier transform Eqs. (25) and (26) and normal ordering, the Hamiltonian decomposes into independent momentum blocks.

It is useful to record the Fourier identities (valid on either momentum grid):

$$\sum_{n=1}^N a_n^\dagger a_{n+1} = \sum_k e^{+i\phi_k} \tilde{a}_k^\dagger \tilde{a}_k, \quad \sum_{n=1}^N a_n a_{n+1} = \sum_k e^{-i\phi_k} \tilde{a}_k \tilde{a}_{-k}, \quad (27)$$

$$\sum_{n=1}^N a_n^\dagger a_{n+1}^\dagger = \sum_k e^{+i\phi_k} \tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger, \quad \sum_{n=1}^N a_n a_{n+1}^\dagger = \sum_k e^{-i\phi_k} \tilde{a}_k \tilde{a}_k^\dagger. \quad (28)$$

For completeness, we show the derivation of the first identity (the rest follow analogously). Using Eqs. (25)

and (26) and the boundary-compatible momentum grid,

$$\begin{aligned}
\sum_{n=1}^N a_n^\dagger a_{n+1} &= \frac{1}{N} \sum_{n=1}^N \sum_{k,k'} e^{-in\phi_k} e^{+i(n+1)\phi_{k'}} \tilde{a}_k^\dagger \tilde{a}_{k'} \\
&= \frac{1}{N} \sum_{k,k'} e^{+i\phi_{k'}} \left(\sum_{n=1}^N e^{in(\phi_{k'} - \phi_k)} \right) \tilde{a}_k^\dagger \tilde{a}_{k'} \\
&= \sum_k e^{+i\phi_k} \tilde{a}_k^\dagger \tilde{a}_k, \tag{29}
\end{aligned}$$

where we used $\sum_{n=1}^N e^{in(\phi_{k'} - \phi_k)} = N\delta_{k,k'}$ on the allowed momentum grid. As a second example (relevant for the pairing sector), we derive $\sum_n a_n a_{n+1} = \sum_k e^{-i\phi_k} \tilde{a}_k \tilde{a}_{-k}$:

$$\begin{aligned}
\sum_{n=1}^N a_n a_{n+1} &= \frac{1}{N} \sum_{n=1}^N \sum_{k,k'} e^{+in\phi_k} e^{+i(n+1)\phi_{k'}} \tilde{a}_k \tilde{a}_{k'} \\
&= \frac{1}{N} \sum_{k,k'} e^{+i\phi_{k'}} \left(\sum_{n=1}^N e^{in(\phi_k + \phi_{k'})} \right) \tilde{a}_k \tilde{a}_{k'} \\
&= \sum_k e^{-i\phi_k} \tilde{a}_k \tilde{a}_{-k}, \tag{30}
\end{aligned}$$

where we used $\sum_{n=1}^N e^{in(\phi_k + \phi_{k'})} = N\delta_{k',-k}$ and $\phi_{-k} = -\phi_k$.

Deriving the $(k, -k)$ block. We now show the intermediate steps leading from the real-space fermion Hamiltonian Eq. (22) to the canonical paired-mode form Eq. (35). First expand the XX part and use the Fourier identities above:

$$\begin{aligned}
H_{\text{spin}}^{g_F=1} &= \frac{\cos \theta}{2} \sum_{n=1}^N (a_n - a_n^\dagger)(a_{n+1} + a_{n+1}^\dagger) - \frac{\sin \theta}{2} \sum_{n=1}^N (a_n a_n^\dagger - a_n^\dagger a_n) \\
&= \frac{\cos \theta}{2} \sum_{n=1}^N (a_n a_{n+1} + a_n a_{n+1}^\dagger - a_n^\dagger a_{n+1} - a_n^\dagger a_{n+1}^\dagger) - \frac{\sin \theta}{2} \sum_{n=1}^N (a_n a_n^\dagger - a_n^\dagger a_n) \\
&= \frac{\cos \theta}{2} \sum_k \left(e^{-i\phi_k} \tilde{a}_k \tilde{a}_{-k} + e^{-i\phi_k} \tilde{a}_k \tilde{a}_k^\dagger - e^{+i\phi_k} \tilde{a}_k^\dagger \tilde{a}_k - e^{+i\phi_k} \tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger \right) - \frac{\sin \theta}{2} \sum_k \left(\tilde{a}_k \tilde{a}_k^\dagger - \tilde{a}_k^\dagger \tilde{a}_k \right). \tag{31}
\end{aligned}$$

Now consider a generic paired mode $k \neq 0, N/2$ and group the k and $-k$ contributions (using $\phi_{-k} = -\phi_k$ on either

momentum grid). It is convenient to separate number and pairing sectors.

$$\begin{aligned}
\mathcal{N}_{k,-k} &\equiv e^{-i\phi_k} \tilde{a}_k \tilde{a}_k^\dagger - e^{+i\phi_k} \tilde{a}_k^\dagger \tilde{a}_k + e^{+i\phi_k} \tilde{a}_{-k} \tilde{a}_{-k}^\dagger - e^{-i\phi_k} \tilde{a}_{-k}^\dagger \tilde{a}_{-k} \\
&= (e^{-i\phi_k} + e^{+i\phi_k}) - (e^{-i\phi_k} + e^{+i\phi_k}) (\tilde{a}_k^\dagger \tilde{a}_k + \tilde{a}_{-k}^\dagger \tilde{a}_{-k}) \\
&= -2 \cos \phi_k (\tilde{a}_k^\dagger \tilde{a}_k + \tilde{a}_{-k}^\dagger \tilde{a}_{-k} - 1), \tag{32}
\end{aligned}$$

where we used normal ordering $\tilde{a}_k \tilde{a}_k^\dagger = 1 - \tilde{a}_k^\dagger \tilde{a}_k$ and

$e^{-i\phi} + e^{+i\phi} = 2 \cos \phi$. Similarly for the pairing sector,

$$\begin{aligned}
\mathcal{P}_{k,-k} &\equiv e^{-i\phi_k} \tilde{a}_k \tilde{a}_{-k} - e^{+i\phi_k} \tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger + e^{+i\phi_k} \tilde{a}_{-k} \tilde{a}_k - e^{-i\phi_k} \tilde{a}_{-k}^\dagger \tilde{a}_k^\dagger \\
&= (e^{-i\phi_k} - e^{+i\phi_k}) (\tilde{a}_k \tilde{a}_{-k} + \tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger) \\
&= -2i \sin \phi_k (\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger - \tilde{a}_{-k} \tilde{a}_k), \tag{33}
\end{aligned}$$

where we used fermionic anticommutation to reorder $\tilde{a}_{-k}\tilde{a}_k = -\tilde{a}_k\tilde{a}_{-k}$ and $\tilde{a}_{-k}^\dagger\tilde{a}_k^\dagger = -\tilde{a}_k^\dagger\tilde{a}_{-k}^\dagger$, and then Eu-

ler's identity $e^{-i\phi} - e^{+i\phi} = -2i \sin \phi$.

Combining the k and $-k$ pieces of Eq. (31) gives

$$\begin{aligned} H_{k,-k} &= \frac{\cos \theta}{2} (\mathcal{N}_{k,-k} + \mathcal{P}_{k,-k}) \\ &\quad - \frac{\sin \theta}{2} [(\tilde{a}_k \tilde{a}_k^\dagger - \tilde{a}_k^\dagger \tilde{a}_k) + (\tilde{a}_{-k} \tilde{a}_{-k}^\dagger - \tilde{a}_{-k}^\dagger \tilde{a}_{-k})] \\ &= (\sin \theta - \cos \theta \cos \phi_k) (\tilde{a}_k^\dagger \tilde{a}_k + \tilde{a}_{-k}^\dagger \tilde{a}_{-k} - 1) \\ &\quad + i(-\cos \theta \sin \phi_k) (\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger - \tilde{a}_{-k} \tilde{a}_k), \end{aligned} \quad (34)$$

which is the canonical form quoted below.

Applying these steps to Eq. (22) and normal ordering $\tilde{a}_k \tilde{a}_k^\dagger = 1 - \tilde{a}_k^\dagger \tilde{a}_k$, the $(k, -k)$ contribution for a generic paired mode ($k \neq 0, N/2$) can be written as

$$H_{k,-k} = w_k (\tilde{a}_k^\dagger \tilde{a}_k + \tilde{a}_{-k}^\dagger \tilde{a}_{-k} - 1) + ir_k (\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger - \tilde{a}_{-k} \tilde{a}_k), \quad (35)$$

with

$$w_k = \sin \theta - \cos \theta \cos \phi_k, \quad r_k = -\cos \theta \sin \phi_k. \quad (36)$$

For the generic paired modes $k \neq 0, N/2$ we define the Nambu spinor $\tilde{\alpha}_k = (\tilde{a}_k, \tilde{a}_{-k}^\dagger)^T$ and obtain

$$H_{\text{spin}}^{g_F=1} = \sum_{k=1}^{N/2-1} \tilde{\alpha}_k^\dagger \tilde{H}_k \tilde{\alpha}_k + H_{k=0} + H_{k=N/2}, \quad (37)$$

where the 2×2 Bogoliubov-de Gennes block is

$$\tilde{H}_k = \begin{pmatrix} w_k & ir_k \\ -ir_k & -w_k \end{pmatrix}, \quad w_k = \sin \theta - \cos \theta \cos \phi_k, \quad r_k = -\cos \theta \sin \phi_k. \quad (38)$$

As a quick consistency check, expanding $\tilde{\alpha}_k^\dagger \tilde{H}_k \tilde{\alpha}_k$ gives

$$\begin{aligned} \tilde{\alpha}_k^\dagger \tilde{H}_k \tilde{\alpha}_k &= w_k (\tilde{a}_k^\dagger \tilde{a}_k - \tilde{a}_{-k} \tilde{a}_{-k}^\dagger) + ir_k (\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger - \tilde{a}_{-k} \tilde{a}_k) \\ &= w_k (\tilde{a}_k^\dagger \tilde{a}_k + \tilde{a}_{-k}^\dagger \tilde{a}_{-k} - 1) + ir_k (\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger - \tilde{a}_{-k} \tilde{a}_k), \end{aligned} \quad (39)$$

which matches Eq. (35) after normal ordering. The special modes have $r_k = 0$ and are already diagonal:

$$H_{k=0} = \frac{w_0}{2} (\tilde{a}_0^\dagger \tilde{a}_0 - \tilde{a}_0 \tilde{a}_0^\dagger), \quad w_0 = \sin \theta - \cos \theta, \quad (40)$$

$$H_{k=N/2} = \frac{w_{N/2}}{2} (\tilde{a}_{N/2}^\dagger \tilde{a}_{N/2} - \tilde{a}_{N/2} \tilde{a}_{N/2}^\dagger), \quad w_{N/2} = \sin \theta + \cos \theta. \quad (41)$$

F. Fermionic APBC $g_F = -1$ Case

For $g_F = -1$ (fermionic APBC) the allowed momenta are half-integers $k \in \{-\frac{N-1}{2}, \dots, \frac{N-1}{2}\}$. There are no

special modes ($\phi_k \notin \{0, \pi\}$), and the Hamiltonian again decomposes into independent Nambu blocks of the form Eq. (38):

$$H_{\text{spin}}^{g_F=-1} = \sum_{k=1/2}^{(N-1)/2} \tilde{\alpha}_k^\dagger \tilde{H}_k \tilde{\alpha}_k. \quad (42)$$

II. BOGOLIUBOV TRANSFORMATION

To fully diagonalize the Hamiltonian and obtain the free quasiparticle energies, we apply a Bogoliubov transformation. For modes with $k \neq 0, N/2$, we need to diagonalize the 2×2 matrix $\tilde{H}_k$. The Bogoliubov transformation mixes the $\tilde{a}_k$ and $\tilde{a}_{-k}^\dagger$ operators:

$$\hat{a}_k = \cos(\varphi_k) \tilde{a}_k + i \sin(\varphi_k) \tilde{a}_{-k}^\dagger, \quad (43)$$

$$\hat{a}_{-k}^\dagger = i \sin(\varphi_k) \tilde{a}_k + \cos(\varphi_k) \tilde{a}_{-k}^\dagger, \quad (44)$$

where the Bogoliubov angle φ_k is chosen to diagonalize $\tilde{H}_k$.

To find the angle φ_k , we need to diagonalize the matrix:

$$\tilde{H}_k = \begin{pmatrix} w_k & ir_k \\ -ir_k & -w_k \end{pmatrix}. \quad (45)$$

The eigenvalues of this matrix are $\pm \epsilon_k$ where:

$$\begin{aligned} \epsilon_k &= \sqrt{w_k^2 + r_k^2} \\ &= \sqrt{(\sin \theta - \cos \theta \cos \phi_k)^2 + (\cos \theta \sin \phi_k)^2} \\ &= \sqrt{\sin^2 \theta - 2 \sin \theta \cos \theta \cos \phi_k + \cos^2 \theta (\cos^2 \phi_k + \sin^2 \phi_k)} \\ &= \sqrt{1 - 2 \sin \theta \cos \theta \cos \phi_k} = \sqrt{1 - \sin(2\theta) \cos \phi_k}. \end{aligned} \quad (46)$$

The Bogoliubov angle is determined by:

$$\tan(2\varphi_k) = \frac{r_k}{w_k} \quad (48)$$

$$= \frac{-\cos \theta \sin \phi_k}{\sin \theta - \cos \theta \cos \phi_k}. \quad (49)$$

Let us verify that this transformation diagonalizes the Hamiltonian. The transformation can be written in matrix form as:

$$\begin{pmatrix} \hat{a}_k \\ \hat{a}_{-k}^\dagger \end{pmatrix} = U_k \begin{pmatrix} \tilde{a}_k \\ \tilde{a}_{-k}^\dagger \end{pmatrix}, \quad U_k = \begin{pmatrix} \cos \varphi_k & i \sin \varphi_k \\ i \sin \varphi_k & \cos \varphi_k \end{pmatrix}. \quad (50)$$

The Hamiltonian matrix for mode k is diagonalized by this transformation. With the correct choice of φ_k , we have:

$$U_k \tilde{H}_k U_k^\dagger = \begin{pmatrix} \epsilon_k & 0 \\ 0 & -\epsilon_k \end{pmatrix}. \quad (51)$$

Therefore, for $k \in \{1, \dots, N/2 - 1\}$:

$$\tilde{\alpha}_k^\dagger \tilde{H}_k \tilde{\alpha}_k = \epsilon_k \hat{a}_k^\dagger \hat{a}_k - \epsilon_k \hat{a}_{-k} \hat{a}_{-k}^\dagger \quad (52)$$

$$= \epsilon_k \hat{a}_k^\dagger \hat{a}_k - \epsilon_k (1 - \hat{a}_{-k}^\dagger \hat{a}_{-k}) \quad (53)$$

$$= \epsilon_k (\hat{a}_k^\dagger \hat{a}_k - \hat{a}_{-k} \hat{a}_{-k}^\dagger). \quad (54)$$

For the special modes $k = 0$ and $k = N/2$ (present only for fermionic PBC, $g_F = +1$), we have $\sin \phi_k = 0$ so $r_k = 0$ and the Hamiltonian is already diagonal:

$$H_{k=0} = \frac{w_0}{2} (\tilde{a}_0^\dagger \tilde{a}_0 - \tilde{a}_0 \tilde{a}_0^\dagger) = \frac{w_0}{2} (\hat{a}_0^\dagger \hat{a}_0 - \hat{a}_0 \hat{a}_0^\dagger), \quad (55)$$

$$H_{k=N/2} = \frac{w_{N/2}}{2} (\tilde{a}_{N/2}^\dagger \tilde{a}_{N/2} - \tilde{a}_{N/2} \tilde{a}_{N/2}^\dagger) = \frac{w_{N/2}}{2} (\hat{a}_{N/2}^\dagger \hat{a}_{N/2} - \hat{a}_{N/2} \hat{a}_{N/2}^\dagger) \quad (56)$$

with $w_0 = \sin \theta - \cos \theta$ and $w_{N/2} = \sin \theta + \cos \theta$. The corresponding excitation energies are $\epsilon_0 = |w_0|$ and $\epsilon_{N/2} = |w_{N/2}|$.

Collecting all modes, define the Bogoliubov number operator

$$\hat{n}_k \equiv \hat{a}_k^\dagger \hat{a}_k, \quad h_k \equiv 2\hat{n}_k - 1. \quad (57)$$

For a generic paired block, Eq. (35) becomes

$$\tilde{\alpha}_k^\dagger \tilde{H}_k \tilde{\alpha}_k = \epsilon_k (\hat{n}_k + \hat{n}_{-k} - 1) = \frac{\epsilon_k}{2} (h_k + h_{-k}). \quad (58)$$

For a special mode $q \in \{0, N/2\}$, present only when $g_F = +1$, the contribution is instead

$$H_q = \frac{w_q}{2} h_q, \quad (59)$$

with the signed coefficient w_q rather than $|w_q|$. Thus the full-grid form used by the numerical diagnostics is

$$H_{\text{spin}}^{g_F} = \frac{1}{2} \sum_{k \in \text{grid}(g_F)} \text{coeff}_k h_k, \quad (60)$$

where

$$\text{coeff}_k = \begin{cases} \epsilon_k, & \sin \phi_k \neq 0, \\ w_k, & \sin \phi_k = 0. \end{cases} \quad (61)$$

In the code parametrization $H_{\text{code}} = J \sum_j \sigma_j^z \sigma_{j+1}^z + h \sum_j \sigma_j^x$, related to this note by the global rotation $R_y(\pi/2)^{\otimes N}$, the energy scale is $\Lambda = 2\sqrt{J^2 + h^2}$ with $\theta = \text{atan}(h, J)$, and the reconstructed energy is

$$\langle H_{\text{code}} \rangle = \frac{\Lambda}{2} \sum_{k \in \text{grid}(g_F)} \text{coeff}_k \langle h_k \rangle. \quad (62)$$

The Bogoliubov quasiparticle occupation reported in the cooling diagnostics is therefore

$$n_k^{\text{Bog}} = \langle \hat{n}_k \rangle = \frac{1 + \langle h_k \rangle}{2}. \quad (63)$$

This occupation should not be confused with the raw Fourier occupation

$$\tilde{n}_k \equiv \langle \tilde{a}_k^\dagger \tilde{a}_k \rangle = \frac{1}{N} \sum_{m,n=1}^N e^{i(m-n)\phi_k} \langle a_m^\dagger a_n \rangle. \quad (64)$$

With the relabeled Fourier sign used in the code one instead writes $\tilde{n}_k^{\text{code}} = \tilde{n}_{-k}$, equivalently replacing $e^{i(m-n)\phi_k}$ in Eq. (64) by $e^{i(n-m)\phi_k}$. This is only a relabeling of the momentum grid, but it must not be hidden in the notation. In the Bogoliubov vacuum, $\tilde{n}_k = \sin^2 \varphi_k$ for generic modes, whereas $n_k^{\text{Bog}} = 0$ and $\langle h_k \rangle = -1$. The energy reconstruction uses h_k and the signed coefficient coeff_k , not the expression $\epsilon_k \tilde{n}_k$. The individual observables h_k and the signed coefficients coeff_k in Eq. (62) are the defining convention for mode-resolved energy diagnostics. The ED and tensor-network implementations are tested against the same spin-state expectation values of these h_k 's, with the special-mode signs kept in coeff_k rather than absorbed into an absolute excitation energy. The separate density-matrix cooling-channel convention is recorded in the tensor-network note; it does not change the Jordan–Wigner formulas above.

A. Going back to the spin operators

To track the cooling of specific modes in the spin model, the ED and tensor-network diagnostics measure the expectation value of the individual mode operators h_k in Eq. (57). Our goal is to express these operators in terms of the original Pauli operators, which involves tracing back through the sequence of transformations (Bogoliubov, Fourier, JW).

1. Generic modes ($k \neq 0, N/2$)

It is useful first to record the paired-block observable

$$\mathfrak{h}_{k,-k} \equiv \frac{h_k + h_{-k}}{2} = \hat{n}_k + \hat{n}_{-k} - 1. \quad (65)$$

Writing

$$\begin{pmatrix} \hat{a}_k \\ \hat{a}_{-k}^\dagger \end{pmatrix} = U_k \begin{pmatrix} \tilde{a}_k \\ \tilde{a}_{-k}^\dagger \end{pmatrix}, \quad U_k = \begin{pmatrix} \cos \varphi_k & i \sin \varphi_k \\ i \sin \varphi_k & \cos \varphi_k \end{pmatrix}, \quad (66)$$

and using the fermionic anticommutation relations, one obtains the symmetric paired-block form

$$\begin{aligned} \mathfrak{h}_{k,-k} &= \cos(2\varphi_k)(\tilde{a}_k^\dagger \tilde{a}_k + \tilde{a}_{-k}^\dagger \tilde{a}_{-k} - 1) \\ &\quad + \sin(2\varphi_k) i(\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger - \tilde{a}_{-k} \tilde{a}_k). \end{aligned} \quad (67)$$

The individual diagnostic stored by the code is instead

$$\begin{aligned} h_k &= 2\hat{a}_k^\dagger \hat{a}_k - 1 \\ &= 2\cos^2 \varphi_k \tilde{a}_k^\dagger \tilde{a}_k + 2\sin^2 \varphi_k (1 - \tilde{a}_{-k}^\dagger \tilde{a}_{-k}) \\ &\quad + 2\sin \varphi_k \cos \varphi_k i(\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger - \tilde{a}_{-k} \tilde{a}_k) - 1. \end{aligned} \quad (68)$$

Now, we substitute the Fourier transforms to express h_k in terms of real-space operators a_n :

$$\tilde{a}_k^\dagger \tilde{a}_k = \frac{1}{N} \sum_{n,m=1}^N e^{i(n-m)\phi_k} a_n^\dagger a_m, \quad (69)$$

$$\tilde{a}_{-k}^\dagger \tilde{a}_{-k} = \frac{1}{N} \sum_{n,m=1}^N e^{i(m-n)\phi_k} a_n^\dagger a_m, \quad (70)$$

$$\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger = \frac{1}{N} \sum_{n,m=1}^N e^{i(n-m)\phi_k} a_n^\dagger a_m^\dagger, \quad (71)$$

$$\tilde{a}_{-k} \tilde{a}_k = \frac{1}{N} \sum_{n,m=1}^N e^{i(n-m)\phi_k} a_n a_m. \quad (72)$$

We next rewrite the JW bilinears in terms of split-string spin operators. Using $\sigma_n^\pm = (\sigma_{n,x} \pm i\sigma_{n,y})/2$ and $a_n = -S_n \sigma_n^-$, we obtain the identities

$$\begin{aligned} a_n^\dagger a_m &= S_n \sigma_n^+ S_m \sigma_m^- = \frac{1}{4} S_n (\sigma_{n,x} + i\sigma_{n,y}) S_m (\sigma_{m,x} - i\sigma_{m,y}) \\ &= \frac{1}{4} (S_n \sigma_{n,x} S_m \sigma_{m,x} + S_n \sigma_{n,y} S_m \sigma_{m,y} + i(S_n \sigma_{n,y} S_m \sigma_{m,x} - S_n \sigma_{n,x} S_m \sigma_{m,y})), \end{aligned} \quad (73)$$

$$\begin{aligned} a_n^\dagger a_m^\dagger &= S_n \sigma_n^+ S_m \sigma_m^+ = \frac{1}{4} S_n (\sigma_{n,x} + i\sigma_{n,y}) S_m (\sigma_{m,x} + i\sigma_{m,y}) \\ &= \frac{1}{4} (S_n \sigma_{n,x} S_m \sigma_{m,x} - S_n \sigma_{n,y} S_m \sigma_{m,y} + i(S_n \sigma_{n,x} S_m \sigma_{m,y} + S_n \sigma_{n,y} S_m \sigma_{m,x})), \end{aligned} \quad (74)$$

$$\begin{aligned} a_n a_m &= S_n \sigma_n^- S_m \sigma_m^- = \frac{1}{4} S_n (\sigma_{n,x} - i\sigma_{n,y}) S_m (\sigma_{m,x} - i\sigma_{m,y}) \\ &= \frac{1}{4} (S_n \sigma_{n,x} S_m \sigma_{m,x} - S_n \sigma_{n,y} S_m \sigma_{m,y} - i(S_n \sigma_{n,x} S_m \sigma_{m,y} + S_n \sigma_{n,y} S_m \sigma_{m,x})). \end{aligned} \quad (75)$$

For compactness in what follows, we denote the split-string two-point spin correlators by

$$C_{nm}^{\alpha\beta} \equiv \langle S_n \sigma_{n,\alpha} S_m \sigma_{m,\beta} \rangle, \quad \alpha, \beta \in \{x, y\}. \quad (76)$$

Third, we compute the individual diagnostic $\langle h_k \rangle$ by combining density and pairing sectors. To this end, define

$$u \equiv \cos \varphi_k, \quad v \equiv \sin \varphi_k. \quad (77)$$

From Eq. (68), we have

$$\begin{aligned} \langle h_k \rangle &= 2u^2 \langle \tilde{a}_k^\dagger \tilde{a}_k \rangle + 2v^2 \langle 1 - \tilde{a}_{-k}^\dagger \tilde{a}_{-k} \rangle \\ &\quad + 2uv \langle i(\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger - \tilde{a}_{-k} \tilde{a}_k) \rangle - 1. \end{aligned} \quad (78)$$

We now evaluate the three expectation values in Eq. (78) explicitly. Introduce the shorthand $\theta_{nm} = (n-m)\phi_k$.

Density sector. Using the Fourier relations in Eqs. (25) and (26), we have

$$\tilde{a}_k^\dagger \tilde{a}_k = \frac{1}{N} \sum_{n,m=1}^N e^{i\theta_{nm}} a_n^\dagger a_m, \quad (79)$$

$$\tilde{a}_{-k}^\dagger \tilde{a}_{-k} = \frac{1}{N} \sum_{n,m=1}^N e^{-i\theta_{nm}} a_n^\dagger a_m. \quad (80)$$

These two terms must be kept separately for the individual observable h_k ; only the paired-block observable $\mathfrak{h}_{k,-k}$ combines them into a cosine.

Pairing sector. Similarly,

$$\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger = \frac{1}{N} \sum_{n,m=1}^N e^{i\theta_{nm}} a_n^\dagger a_m^\dagger, \quad (81)$$

$$\tilde{a}_{-k} \tilde{a}_k = \frac{1}{N} \sum_{n,m=1}^N e^{i\theta_{nm}} a_n a_m, \quad (82)$$

so that

$$\langle i(\tilde{a}_k^\dagger \tilde{a}_{-k}^\dagger - \tilde{a}_{-k} \tilde{a}_k) \rangle = \frac{i}{N} \sum_{n,m=1}^N e^{i\theta_{nm}} \langle a_n^\dagger a_m^\dagger - a_n a_m \rangle. \quad (83)$$

We next expand the real-space fermion bilinears using the split-string correlators $C_{nm}^{\alpha\beta} = \langle S_n \sigma_{n,\alpha} S_m \sigma_{m,\beta} \rangle$. From Eqs. (73) to (75), we have

$$\langle a_n^\dagger a_m \rangle = \frac{1}{4} (C_{nm}^{xx} + C_{nm}^{yy} + i[C_{nm}^{yx} - C_{nm}^{xy}]), \quad (84)$$

$$\langle a_n^\dagger a_m^\dagger - a_n a_m \rangle = \frac{i}{2} (C_{nm}^{xy} + C_{nm}^{yx}). \quad (85)$$

The second identity follows by subtracting Eq. (75) from Eq. (74).

As a useful diagonal check, for $n = m$ we have $S_n^2 = \mathbb{1}$ and $\sigma_{n,\alpha}^2 = \mathbb{1}$, hence $C_{nn}^{xx} = C_{nn}^{yy} = 1$, while the Pauli algebra gives $i[C_{nn}^{yx} - C_{nn}^{xy}] = 2\langle \sigma_{n,z} \rangle$. Thus $\langle a_n^\dagger a_n \rangle = \frac{1}{2}(1 + \langle \sigma_{n,z} \rangle)$ is recovered automatically from the compact correlator combination above.

Combining these identities yields the explicit result

$$\begin{aligned} \langle h_k \rangle &= \frac{u^2}{2N} \sum_{n,m=1}^N e^{i\theta_{nm}} (C_{nm}^{xx} + C_{nm}^{yy} + i[C_{nm}^{yx} - C_{nm}^{xy}]) \\ &\quad - \frac{v^2}{2N} \sum_{n,m=1}^N e^{-i\theta_{nm}} (C_{nm}^{xx} + C_{nm}^{yy} + i[C_{nm}^{yx} - C_{nm}^{xy}]) \\ &\quad - \frac{uv}{N} \sum_{n,m=1}^N e^{i\theta_{nm}} (C_{nm}^{xy} + C_{nm}^{yx}) + 2v^2 - 1. \end{aligned} \quad (86)$$

The right-hand side is real, since h_k is Hermitian. Any nonzero imaginary part provides a useful numerical check of correlator conventions and implementations. Note that the mixed correlators C^{xy} and C^{yx} are generally non-Hermitian, so they must be evaluated with the split-string order as written for each pair (n, m) (do not fill them by conjugate mirroring unless one also swaps operators appropriately). The ED implementation uses the opposite Fourier sign for the intermediate operator $\tilde{a}_k$, which is the relabeling $k \mapsto -k$; the tensor-network implementation uses the same relabeled phases and the same individual observable Eq. (57).

2. Special modes ($k = 0, N/2$)

Special modes occur only for fermionic PBC ($g_F = +1$), where the momentum grid contains $k = 0$ and $k = N/2$ (i.e., $\phi_k \in \{0, \pi\}$). For these modes $\sin \phi_k = 0$ so $r_k = 0$ and there is no Bogoliubov mixing; we set $\varphi_k = 0$ and identify $\hat{a}_k = \tilde{a}_k$. The Hamiltonian contribution is

$$H_k = \frac{w_k}{2} h_k, \quad h_k = 2\tilde{a}_k^\dagger \tilde{a}_k - 1, \quad (87)$$

with $w_0 = \sin \theta - \cos \theta$ and $w_{N/2} = \sin \theta + \cos \theta$ and excitation energies $\epsilon_k = |w_k|$. In a fixed fermion-parity sector, the special-mode occupations are constrained. In any

sector where a single special-mode occupation is forbidden, that mode can remain at $\langle h_k \rangle = -1$ even when w_0 changes sign at $\theta = \pi/4$; the sign change is then carried by the *signed* coefficient w_k in Eq. (60), not by replacing w_k with $|w_k|$.

In terms of split-string correlators, one may obtain $\langle h_k \rangle$ by taking $v = 0$ in Eq. (86). Since $e^{i\theta_{nm}} = \cos \theta_{nm}$ for $\phi_k \in \{0, \pi\}$, this reduces to

$$\begin{aligned} \langle h_k \rangle &= \frac{1}{2N} \sum_{n,m=1}^N \cos \theta_{nm} (C_{nm}^{xx} + C_{nm}^{yy}) \\ &\quad + \frac{i}{2N} \sum_{n,m=1}^N \cos \theta_{nm} (C_{nm}^{yx} - C_{nm}^{xy}) - 1, \end{aligned} \quad (88)$$

where $\theta_{nm} = (n - m)\phi_k$.

B. Ground State Parity and Gap Structure

The parity operator $P = \prod_{j=1}^N \sigma_{j,z}$ commutes with H_{spin} , splitting the Hilbert space into even ($P = +1$) and odd ($P = -1$) sectors. From the mapping $g_F = -g_I \cdot P$, the spin PBC ($g_I = 1$) yields fermionic APBC ($g_F = -1$) in the even sector and fermionic PBC ($g_F = +1$) in the odd sector.

More generally, in a fixed parity sector $P = \pm 1$ the fermionic boundary condition is $g_F = -g_I P$ (equivalently, $g_F = -g_I$ in the even sector and $g_F = +g_I$ in the odd sector), which fixes the allowed momentum grid. In each such sector the Hamiltonian can be written in the full-grid form

$$H_{\text{spin}}^{g_F} = \frac{1}{2} \sum_{k \in \text{grid}(g_F)} \text{coeff}_k h_k, \quad (89)$$

with $\text{coeff}_k = \epsilon_k$ for generic modes and $\text{coeff}_k = w_k$ (signed) for the special modes (when present). The Bogoliubov vacuum, defined by

$$\hat{a}_k |0\rangle = 0 \quad \text{for all } k, \quad (90)$$

has $\langle h_k \rangle = -1$ for all modes (with the special modes unoccupied), and therefore its energy is

$$E_{\text{vac}}^{g_F} = -\frac{1}{2} \sum_{k \in \text{grid}(g_F)} \text{coeff}_k. \quad (91)$$

For $g_F = -1$ (half-integer k grid) there are no special modes and this reduces to $E_{\text{vac}}^{g_F=-1} = -\sum_{k=1/2}^{(N-1)/2} \epsilon_k$. For $g_F = +1$ (integer k grid) the special modes contribute $\text{coeff}_0 = w_0$ and $\text{coeff}_{N/2} = w_{N/2}$ (signed). In a fixed parity sector the true sector ground state can differ from the Bogoliubov vacuum when the allowed fermion parity requires occupying one or more modes; this distinction is tested against exact diagonalization in the code.

Let E_0^{even} and E_0^{odd} denote the lowest energies in the even and odd parity sectors, respectively. There are two distinct energy gaps:

$$\Delta_{\text{intra}} = 2\epsilon_{\text{min}}, \quad (92)$$

$$\Delta_{\text{inter}} = |E_0^{\text{odd}} - E_0^{\text{even}}|. \quad (93)$$

Here Δ_{intra} is the parity-preserving excitation gap within a sector, Δ_{inter} is the parity-changing gap, and ϵ_{min} is the smallest generic mode energy ϵ_k on the corresponding momentum grid.

The hierarchy $\Delta_{\text{inter}} < \Delta_{\text{intra}}$ arises when the energy difference between filled modes in different parity sectors is smaller than creating a particle-hole pair within a sector. For parity-preserving dynamics, excitations are confined within sectors, making Δ_{intra} the relevant energy scale.

This distinction is important for the tensor-network cooling diagnostics. The stored array `mode_ek_values` contains the single-quasiparticle energies ϵ_k in code units. By contrast, the automatic bath-detuning reference for a parity-preserving local system coupling is the many-body scale $\Delta_{\text{intra}} = 2\epsilon_{\text{min}}$, because such a coupling can change the generic Bogoliubov occupations only in pairs within a fixed parity sector. Thus one should not use the minimum of `mode_ek_values` itself as the default detuning for the parity-preserving Ising cooling channel. On a Fourier grid containing special modes, lower fixed-parity transitions involving those special modes may occur; such cases should use an explicit detuning interval until the special-mode transition scale is selected separately.

C. Parity-preserving couplings in the code convention

In the notes basis the Ising parity is

$$P = \prod_{n=1}^N \sigma_{n,z}, \quad (94)$$

and it commutes with the spin Hamiltonian H_{spin} . In the Julia code the Ising Hamiltonian is written in the globally rotated basis $H_{\text{code}} = J \sum_j \sigma_j^z \sigma_{j+1}^z + h \sum_j \sigma_j^x$, so this same parity is represented by

$$P_x = \prod_{n=1}^N \sigma_{n,x}. \quad (95)$$

Thus a local code-basis system Pauli preserves the Ising parity precisely when it is σ_x .

The implemented system-bath coupling is not encoded by a separate fixed bath example. It is specified by the shared two-letter coupling label $c \in \{X, Y, Z\}^2$, whose source of truth is `src/coupling_utils.jl`. The local terms are

$$\mathcal{T}(AA) = \{(A, A)\}, \quad \mathcal{T}(AB) = \{(A, B), (B, A)\} \quad (A \neq B), \quad (96)$$

$$V_{SB}(c) = g \sum_{i=1}^N \sum_{(\mu, \nu) \in \mathcal{T}(c)} \sigma_{s_i, \text{code}}^\mu \sigma_{b_i}^\nu. \quad (97)$$

The bath Hamiltonian and reset state use the same convention as the ED, MPS, and MPO implementations:

$$H_B(c, \Delta) = \frac{\Delta}{2} \sum_{i=1}^N \sigma_{b_i}^{\beta(c)}, \quad |\Omega_B(c)\rangle = \bigotimes_{i=1}^N |\beta(c), -\rangle_{b_i}, \quad (98)$$

where $\beta(c)$ is the Pauli label returned by `get_bath_operator` and $|\beta(c), -\rangle$ is its eigenvalue -1 state. This is the convention used by `construct_system_bath_hamiltonian` and by the tensor-network note.

Consequently the code-basis coupling c preserves the Ising parity if and only if every system-side label μ appearing in $\mathcal{T}(c)$ is X . The large- N mode-detuning driver checks this condition through `campaign_mode_detuning_preserves_px`. For example, XX preserves P_x , while a mixed symmetric label such as XY or XZ does not, because the Hermitian convention also includes a term with a non- X system Pauli. The automatic parity-preserving detuning reference $\Delta_{\text{intra}} = 2\epsilon_{\text{min}}$ should therefore be used only when this condition holds; otherwise the physical detuning interval should be specified explicitly.

III. NUMERICAL SIMULATION OF COOLING DYNAMICS

A. Split-Operator Trotter Method

For the Hamiltonians we are considering, a very effective approach is the split-operator method based on a Trotter-Suzuki decomposition. The total Hamiltonian is $H = H_S + H_B + V$. We can split the total Hamiltonian into two parts that are each easy to exponentiate: $H = H_z + H_x$, where H_z contains all σ_z -type terms (e.g., $\sigma_z \sigma_z$, σ_z) and H_x contains all σ_x -type terms.

$$e^{-iHt} \approx (e^{-iH_z \delta t} e^{-iH_x \delta t})^M, \quad \text{where } t = M\delta t. \quad (99)$$

This is very efficient to apply to a state vector or density matrix:

- $e^{-iH_z \delta t}$ is a diagonal operator in the computational (σ_z) basis. Applying it is just an element-wise multiplication of the state vector by a precomputed vector of phases. This is an $\mathcal{O}(d^N)$ operation.
- $e^{-iH_x \delta t}$ is a product of single-qubit rotations. In the computational basis, this corresponds to applying a tensor product of single-qubit gates, which can also be implemented efficiently.

This method avoids matrix-matrix or matrix-vector multiplication with large, dense matrices and scales much better. A Trotter step of $\delta t = 0.1$ is typically accurate enough.

B. Partial Trace for Density Matrices

In the simulation of the cooling dynamics, we need to trace out bath qubits from a density matrix ρ represented in the computational basis. Tracing out the k -th qubit corresponds to summing subblocks of the density matrix. For example, to trace out the first qubit, you partition ρ into four blocks based on the state of the first qubit:

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}. \quad (100)$$

The traced-out matrix is $\rho' = \rho_{00} + \rho_{11}$. This can be done iteratively for all bath qubits. For large numbers of bath qubits, this can be implemented without loops using clever indexing in MATLAB or Python, by identifying the indices that need to be summed together.

C. Sampling for Quantum Trajectories

For the Monte Carlo wave function method, each cooling cycle involves the following steps:

1. Evolve the joint pure state: $|\Psi'_{SB}\rangle = e^{-iHt}(|\Psi_S\rangle \otimes |0_B\rangle)$.

2. Compute the probability distribution for measuring the bath in each computational basis state $|n\rangle_B$:

$$p_n = \langle \Psi'_{SB} | (\mathbb{1}_S \otimes |n\rangle_B \langle n|) | \Psi'_{SB} \rangle. \quad (101)$$

This vector of 2^{N_B} probabilities can be computed efficiently by taking the diagonal elements of the bath's reduced density matrix, $\rho_B = \text{tr}_S(|\Psi'_{SB}\rangle \langle \Psi'_{SB}|)$.

3. Sample a specific outcome $\tilde{n}$ from the distribution $\{p_n\}$. A standard way is to compute the cumulative sum of the probabilities, $C_k = \sum_{j=1}^k p_j$, draw a single random number $r \in [0, 1]$, and find the first k such that $r \leq C_k$.
4. The new system state is the postmeasurement state, properly normalized:

$$|\Psi_S^{\text{new}}\rangle = \frac{(\mathbb{1}_S \otimes \langle \tilde{n}|_B) |\Psi'_{SB}\rangle}{\sqrt{p_{\tilde{n}}}}. \quad (102)$$